

Distributed Metasolver



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Abstract

Many modern applications, such as artificial intelligence, machine learning, or medical informatics rely on specialized computational models that are optimized using solver software.

The aim of this thesis was to create a distributed metasolver utilizing the island-based model of work distribution, where each island represents a single worker node in a distributed system, with a centralized element being responsible for coordination tasks such as allocating work or distributing optimization problems.

The distributed metasolver was implemented as a C++ library and its performance was evaluated on several standardized optimization problems.

The results have demonstrated satisfactory performance regarding reliability and distribution of work, although the load-balancing mechanism proved to be less efficient in certain scenarios.

Introduction

The sequential paradigm of computing prevailed for most of the 20th century, however new ways of accelerating the execution time of computer programs soon began to be developed.

This started with symmetric multiprocessor (SMP) systems and instruction pipelining in the 1960s, followed by SIMD vector processing in the 1970s.

One of the more recent breakthroughs in

terms of accelerating the execution time of a program, made possible by the advancement of TCP/IP networking in the 1980s and 1990s, was **distributed computing**.

This allowed programs such as the distributed metasolver developed in this thesis to distribute a workload among many nodes in a cluster, efficiently utilizing their combined computational resources.

Population-Based Algorithms

Traditional optimization algorithms, such as gradient descent, maintain a **single** candidate solution which is iteratively improved until an optimum of the optimized function is reached.

This thesis utilizes population-based optimization algorithms, which differ from classical methods in the fact that they utilize a “population” of candidate solutions of an optimization problem, rather than trying to iteratively improve a single one.

These algorithms are especially beneficial for parallel and distributed implementations, as different portions of the population can be optimized independently on multiple processing elements or nodes.

Implementation

The most important classes of the implemented library are *distributed_controller*, *distributed_worker* and *distributed_island*.

- *distributed_controller*
 - Is a network server to which workers connect, it internally uses *ZeroMQ*

sockets and a custom protocol built on top of *ZeroMQ*’s *multi-part messages* feature. The controller’s primary task is to allocate work to available workers and collect their results.

- *distributed_worker*
 - The worker acts as a network client, it receives a population and an optimization problem (distributed as a *DLL* file), the population is evolved and returned back to the controller. Various reliability mechanisms such as heartbeat messages or reconnect interval are included.
- *distributed_island*
 - Implements the Pagmo UDI (User-Defined-Island) interface. Allowing it to be used inside *pagmo::archipelago*.

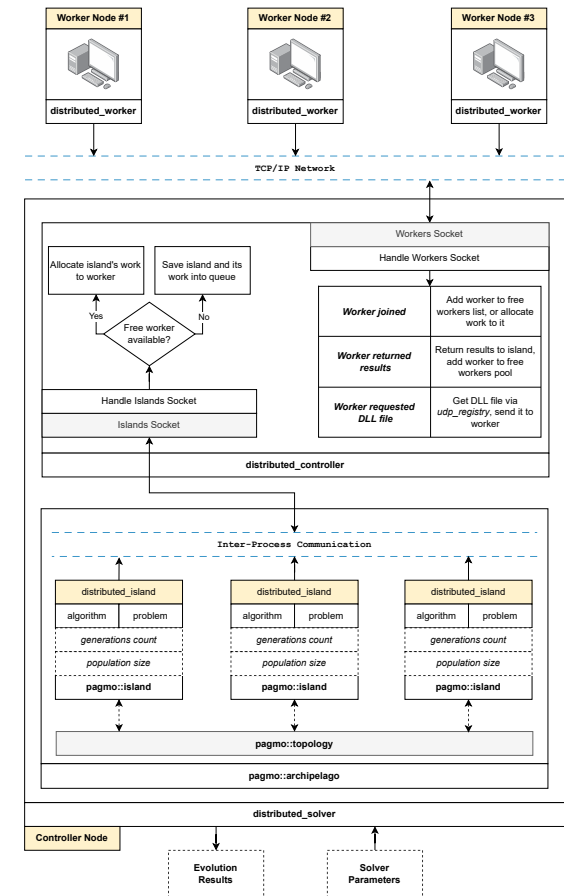
Results and Conclusion

The performance characteristics of this library were evaluated on several testing scenarios.

In a single-algorithm scenario, the results have shown a 115% speed increase and 24% better solution fitness using a two-node distributed cluster compared to local execution.

However, some metasolving scenarios underperformed due to the large difference in computational complexity between the used algorithms.

Future work could include for example improving the Pagmo library’s capabilities, or implementing more robust load-balancing.



Architecture of the metasolver distributed system, showcasing three connected workers.